Algebra 2 — Unit 3: Polynomial Functions

Practice Test — Study Copy — Answer Key

Name: ______________________ Date: _____________ Period: _______

Keep in Mind

This is a practice test. It mirrors the real Unit 3 test in length, format, and the ideas it covers, but the numbers and contexts differ. Work every problem with no notes, then check your answers against the key.

Part A — Vocabulary (8 pts)

Write the letter of the best description next to each term.

1. Degree **B**
2. Leading coefficient **A**
3. End behavior **E**
4. Multiplicity of a zero **C**
5. Turning point **D**
6. Factor Theorem **F**
7. Rational Root Theorem **G**
8. Fundamental Theorem of Algebra **H**

(A) The number multiplying the highest-power term; its sign decides the right-hand arm.

(B) The greatest exponent on the variable once the polynomial is in standard form.

(C) How many times the factor $(x-r)$ appears — odd means the graph crosses at r, even means it touches.

(D) A point where the graph changes from increasing to decreasing, or the reverse.

(E) What the graph does at its far left and far right, as $x \to \pm\infty$.

(F) $(x-r)$ is a factor of $f(x)$ exactly when $f(r) = 0$.

(G) Every rational zero of a polynomial with integer coefficients looks like $\pm\frac{p}{q}$, where p divides the constant term and q divides the leading coefficient.

(H) A polynomial of degree $n \geq 1$ has exactly n complex zeros, counted with multiplicity.

Part B — Multiple Choice (12 pts)

Circle the best answer.

1. The end behavior of $f(x) = -2x^4 + 3x^2 - 1$ is (A) both arms up **(B) both arms down** ← (C) left down, right up (D) left up, right down
2. $(2x-5)^2 =$ (A) $4x^2 - 25$ (B) $4x^2 + 25$ **(C)** $4x^2 - 20x + 25$ ← (D) $4x^2 - 10x + 25$
3. Factored completely, $x^3 + 27$ is **(A)** $(x+3)(x^2-3x+9)$ ← (B) $(x+3)(x^2+3x+9)$ (C) $(x+3)^3$ (D) prime
4. The remainder when $f(x) = x^3 - 4x^2 + 1$ is divided by $x - 2$ is **(A)** -7 ← (B) -23 (C) 7 (D) 0
5. Which value is *not* a possible rational zero of $f(x) = 3x^3 + 2x^2 - x + 4$? (A) -2 (B) $\frac{4}{3}$ **(C)** $\frac{3}{4}$ ← (D) 1
6. A polynomial with *real* coefficients has degree 4 and zeros 2, -1, and $3i$. Its fourth zero must be **(A)** $-3i$ ← (B) $3i$ (C) -3 (D) $\frac{1}{3}$

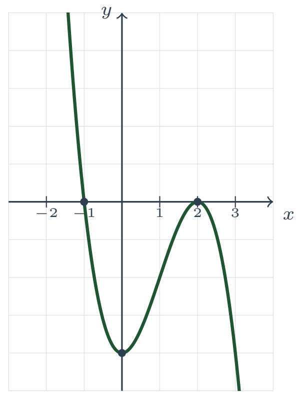
Teacher Note

Item 1: even degree ⇒ the arms match; $a = -2 < 0$ ⇒ they both point down. Only two facts are needed — the $3x^2$ and the -1 are irrelevant. Item 2: (A) and (B) are the $(a-b)^2 = a^2 - b^2$ error, (D) forgets to double the middle product. Item 4: Remainder Theorem, $f(2) = 8 - 16 + 1 = -7$; (B) is $f(-2)$ — the sign-flip error. Item 5: $p \mid 4$ and $q \mid 3$, so the list is $\pm1, \pm2, \pm4, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}$. $\frac{3}{4}$ is the *flipped* fraction. Item 6: real coefficients force conjugate pairs; $3i$ cannot travel alone.

Part C — Short Answer & Computation (40 pts)

Show your work.

1. **Read the graph.** The polynomial function f is graphed below. Each gridline is one unit.



   (a) Degree: even or odd? **odd** Leading coefficient: positive or negative? **negative**

   (b) As $x \to -\infty$, $f(x) \to +\infty$; as $x \to +\infty$, $f(x) \to -\infty$

   (c) Zeros, each with its multiplicity, and whether the graph crosses or touches there: $x = -1$ **(multiplicity 1, crosses);** $x = 2$ **(multiplicity 2, touches)**

   (d) y-intercept: $(0, -4)$; number of turning points: **2**

   (e) Relative maximum: $(2, 0)$; relative minimum: $(0, -4)$;

   absolute maximum or minimum: **none — odd degree, so the arms run to** $\pm\infty$

   (f) Interval(s) where f is increasing: $(0, 2)$; domain: **all reals**; range: **all reals**

2. **Operations.** Simplify. Write each answer in standard form.

   (a) $(3x^3 - 5x + 2) - (x^3 + 4x^2 - 5) = 3x^3 - 5x + 2 - x^3 - 4x^2 + 5 = 2x^3 - 4x^2 - 5x + 7$. **(The subtraction hits *every* term of the second polynomial:** -5 **becomes** $+5$**.)**
   (b) $(2x - 3)(x^2 + 4x - 1) = 2x^3 + 8x^2 - 2x - 3x^2 - 12x + 3 = 2x^3 + 5x^2 - 14x + 3$
   (c) For your answer to (b), give the degree **3**, the leading coefficient **2**, and the end behavior **left down, right up**

3. **Factor completely.** Every factor must be prime over the integers.

   (a) $2x^3 - 18x$ (b) $27x^3 - 8$ (c) $x^3 + 3x^2 - 4x - 12$

   (a) GCF first: $2x(x^2 - 9) = 2x(x - 3)(x + 3)$.

   (b) Difference of cubes with $a = 3x$, $b = 2$: $(3x - 2)\big((3x)^2 + (3x)(2) + 2^2\big) = (3x - 2)(9x^2 + 6x + 4)$; **the quadratic factor is prime.**

   (c) Four terms ⇒ grouping: $x^2(x + 3) - 4(x + 3) = (x + 3)(x^2 - 4) = (x + 3)(x - 2)(x + 2)$.

4. **Division and the Remainder Theorem.**

   (a) Divide using *long division* and write the answer as $Q(x) + \dfrac{R}{D(x)}$:

$$(x^3 + 2x - 9) \div (x - 2)$$

Write the dividend with a placeholder: x^3+0x^2+2x-9**. Long division gives** $Q(x)=x^2+2x+6$ **with** $R=3$**, so the answer is** $x^2+2x+6+\dfrac{3}{x-2}$**. Check:** $(x-2)(x^2+2x+6)+3=x^3+2x-12+3=x^3+2x-9$ ✓

(b) Use the *Remainder Theorem* to find the remainder when $g(x)=x^3+2x^2-5x+6$ is divided by $x+3$. Is $x+3$ a factor of g? Explain in one sentence. $x+3=x-(-3)$**, so the remainder is** $g(-3)=-27+18+15+6=12$**. Since the remainder is** 12 **and not** 0**,** $x+3$ **is *not* a factor (the Factor Theorem requires** $g(-3)=0$**).**

5. **Synthetic division.** You are told that $x=3$ is a zero of

$$f(x)=x^3-2x^2-5x+6.$$

Use synthetic division to divide out that zero, factor f completely, and list *all* of its zeros. **Synthetic division by** $r=3$ **on** $1,\ -2,\ -5,\ 6$ **gives the bottom row** $1,\ 1,\ -2,\ 0$ **— remainder** 0**, confirming the zero. The depressed polynomial is** $x^2+x-2=(x+2)(x-1)$**, so** $f(x)=(x-3)(x+2)(x-1)$ **and the zeros are** $x=3,\ x=-2,\ x=1$**.**

6. **Rational Root Theorem.** For $f(x)=2x^3-x^2-8x+4$:

(a) List every possible rational zero. $p\mid 4$ **gives** $p=\pm1,\pm2,\pm4$**;** $q\mid 2$ **gives** $q=\pm1,\pm2$**. The list (reduced, no repeats) is** $\pm1,\ \pm2,\ \pm4,\ \pm\frac{1}{2}$**.**

(b) Find all real zeros of f, showing the test and the division you used. **Test** $x=2$**:** $f(2)=16-4-16+4=0$**, so** $x=2$ **is a zero. Synthetic division by** 2 **on** $2,\ -1,\ -8,\ 4$ **gives** $2,\ 3,\ -2,\ 0$**, so the depressed polynomial is** $2x^2+3x-2=(2x-1)(x+2)$**. Hence** $f(x)=(x-2)(2x-1)(x+2)$ **and the zeros are** $x=2,\ x=\frac{1}{2},\ x=-2$**. Note** $(2x-1)$ **gives** $x=\frac{1}{2}$ **— not** 2**.**

7. **Solve over the complex numbers.** $x^3-3x^2+4x-12=0$

(a) Before solving: how many solutions must this equation have, counted with multiplicity, and which theorem says so? **Exactly** 3**, by the Fundamental Theorem of Algebra**

(b) Solve. Give every solution. **Group:** $x^2(x-3)+4(x-3)=(x-3)(x^2+4)=0$**. So** $x=3$**, or** $x^2=-4\Rightarrow x=\pm2i$**. The solutions are** $x=3,\ 2i,\ -2i$ **— three of them, as (a) promised.**

(c) How many of your solutions are real, and how many x-intercepts does the graph of $y=x^3-3x^2+4x-12$ therefore have? **One real solution (**$x=3$**), so exactly one** x**-intercept — a graph can never show the imaginary pair**

8. **Build a polynomial from its zeros.** Write a polynomial function of *least degree* with *real* coefficients whose zeros are -1 and $2i$. Give your answer in factored form *and* in standard form. **Real coefficients force the conjugate** $-2i$ **to be a zero too, so there are three zeros and the least degree is** 3**. Factored:** $f(x)=(x+1)(x-2i)(x+2i)=(x+1)(x^2+4)$**. Standard:** $f(x)=x^3+x^2+4x+4$**.**

Teacher Note

Item 1: (e) is the Unit 3 habit-breaker — $(2,0)$ is a *relative* max only, and an odd-degree polynomial has *no* absolute extrema; the range is all reals for the same reason. Accept interval or inequality notation throughout. Item 1(c): full credit requires the multiplicity *and* the cross/touch, since the touch at $x=2$ is the only evidence of the squared factor. Item 3: no credit for $(x+3)(x^2-4)$ or $2x(x^2-9)$ — "completely" is the standard. Item 4(a): the placeholder $0x^2$ is the graded step; a quotient of degree 1 means it was skipped. Item 6(b): accept a graph-based choice of which candidate to test, but the division must appear. Item 7(b): watch for $\pm$ dropped on the square root, and for "no real solutions" written as the final answer — the question says *over* $\mathbb{C}$.

Part D — Extended Response (12 pts)

Explain your reasoning in full sentences.

1. **The graph plan.** Consider $f(x)=-2(x+2)(x-1)^2(x-3)$. Do *not* expand it. **(a)** State the degree and the leading coefficient, and explain how you got each one from the factored form. What end behavior do they force? **(b)** List the zeros with their multiplicities, and say whether the graph crosses or touches the x-axis at each. **(c)** Find the y-intercept. **(d)** Build a sign chart and state every interval on which $f(x)>0$. **(e)** At most how many

turning points can this graph have, and why? **(a) Degrees add: $1+2+1=4$, so the degree is 4. Leading coefficients multiply: $-2\cdot 1\cdot 1\cdot 1=-2$. An even degree with a negative leading coefficient sends *both* arms down: $f(x)\to-\infty$ at each end.**

(b) $x=-2$ (multiplicity 1, crosses); $x=1$ (multiplicity 2, touches); $x=3$ (multiplicity 1, crosses).

(c) $f(0)=-2(2)(-1)^2(-3)=12$, so the y-intercept is $(0,12)$.

(d) Test one value in each interval, keeping only signs: $f(-3)<0$, $f(0)=12>0$, $f(2)=8>0$, $f(4)<0$. So $f(x)>0$ on $(-2,1)\cup(1,3)$ — that is, for $-2<x<3$ except $x=1$ — and $f(x)<0$ on $(-\infty,-2)\cup(3,\infty)$. The sign does *not* flip at $x=1$; that is the fingerprint of the even multiplicity.

(e) At most $4-1=3$ turning points, since a degree-n polynomial turns at most $n-1$ times.

2. **Modeling.** An open-top box is made from a 12 in by 6 in sheet of cardboard by cutting a square of side x from each corner and folding up the sides. **(a)** Write the volume $V(x)$ in factored form, then expand it into standard form. **(b)** State the feasible domain of the model and justify it from the sheet's dimensions. **(c)** Find $V(1)$ and state its units. **(d)** Check that $x=6$ makes $V(x)=0$. Explain why 6 is a genuine zero of the polynomial but not a possible cut length. **(a) Cutting x from both ends of each side leaves a base of $(12-2x)$ by $(6-2x)$ and a height of x, so $V(x)=x(12-2x)(6-2x)$. Expanding: $(12-2x)(6-2x)=4x^2-36x+72$, so $V(x)=4x^3-36x^2+72x$.**

   (b) Every dimension must be positive: $x>0$, $12-2x>0$ (so $x<6$), and $6-2x>0$ (so $x<3$). The tightest of these wins, so the feasible domain is $0<x<3$ — the short side is what limits the cut.

   (c) $V(1)=1(10)(4)=40$ cubic inches (in^3). Check against standard form: $4-36+72=40$ ✓

   (d) $V(6)=6(12-12)(6-12)=6\cdot 0\cdot(-6)=0$, so 6 really is a zero of the polynomial. But a 6-inch cut from a 6-inch side removes the entire width, and $6-2(6)=-6$ is a negative length — there is no box at all. The polynomial does not know about cardboard; only the feasible domain $0<x<3$ describes a real box.

Teacher Note

Scoring Part D (6 pts each). D1: 1 pt (a) degree 4 *and* lead -2 with both arms down; 2 pts (b) all three zeros with multiplicities and cross/touch; 1 pt (c) $(0,12)$; 1 pt (d) $(-2,1)\cup(1,3)$ with the no-flip at $x=1$ noted; 1 pt (e) 3, justified by $n-1$. Give no credit in (a) for "degree 3" from counting factors — the exponents must be added. D2: 2 pts (a) factored *and* expanded; 2 pts (b) $0<x<3$ justified by $6-2x>0$ (a domain of $0<x<6$ earns 1); 1 pt (c) 40 in^3 with cubic units; 1 pt (d) $x=6$ recognized as a real zero of V but rejected because the width $6-2x$ would be negative. Do not accept "because it is too big" without the dimension.